Definition

A *symmetric stacking category* is an associative stacking category $\langle \mathbf{C}, \otimes, \otimes \rangle$ with

<u>Constituents</u>

1. For any two objects $X, Y \in \mathrm{Ob}_{\mathbf{C}}$ there exists an isomorphism

$$\sigma_{X,Y} : X \otimes Y \to Y \otimes X,$$

   called the *braiding*.

<u>Conditions</u>

1. *Naturality*: For any morphisms $f : X \to Z$, $g : Y \to U$, the diagram

$$\begin{tikzcd}
X \otimes Y \arrow[r, "\sigma_{X,Y}"] \arrow[d, "f \otimes g"'] & Y \otimes X \arrow[d, "g \otimes f"] \\
Z \otimes U \arrow[r, "\sigma_{Z,U}"'] & U \otimes Z
\end{tikzcd}$$

   commutes.

2. *Compatibility with nesting*:

$$\begin{tikzcd}
 & (X \otimes Y) \otimes Z \arrow[dl, equal] \arrow[dr, "\sigma_{X,Y} \otimes \mathrm{id}_Z"] & \\
X \otimes (Y \otimes Z) \arrow[d, "\sigma_{X,Y \otimes Z}"] & & (Y \otimes X) \otimes Z \arrow[d, equal] \\
(Y \otimes Z) \otimes X \arrow[dr, equal] & & Y \otimes (X \otimes Z) \arrow[dl, "\mathrm{id}_Y \otimes \sigma_{X,Z}"] \\
 & Y \otimes (Z \otimes X) &
\end{tikzcd}$$

3. *Symmetry*: For all $X, Y \in \mathrm{Ob}_{\mathbf{C}}$,

$$\sigma_{X,Y} \fatsemi \sigma_{Y,X} = \mathrm{id}_{X \otimes Y}.$$